Robust Q-Learning under Corrupted Rewards

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Problem and method. A synchronous generative model supplies observations for every state-action pair. An omniscient adversary may replace reward sets in at most an ϵ fraction of every time prefix. **ϵ-Robust Q-Learning** uses the history of rewards, a robust trimmed mean, and dynamic clipping to keep Q-updates stable under arbitrary reward replacements.

Assumptions and notation. The MDP is finite with S states, A actions, and $\gamma \in (0,1)$. Clean random rewards satisfy $|r(s,a)| \le \bar{r}$, where $\bar{r} \ge 1$. At every iteration, the learner receives an independent successor-state sample for each pair; these transitions remain clean. Let $e_t := \|Q_t - Q^\star\|_\infty$. The robust theorem assumes $\epsilon \in [0, 1/16)$.

Robust update. For each pair, filter its complete reward history and clip the result before the synchronous Q-update:

$$\widehat{r}_t(s,a) = \mathrm{TRIM}\big(\{y_k(s,a)\}_{k=0}^{t}, \epsilon, \delta_1\big), \qquad \widetilde{r}_t(s,a) = \mathrm{clip}_{[-G_t,G_t]}(\widehat{r}_t(s,a)), \qquad \delta_1 = \frac{\delta}{2SAT}.$$

$$Q_{t+1}(s,a) = (1-\alpha)Q_t(s,a) + \alpha\big[\widetilde{r}_t(s,a) + \gamma \max_{a'} Q_t(s'_t(s,a),a')\big].$$

1. Vanilla Q-learning can converge arbitrarily far from the optimum. Under Huber reward injections with mean attack signal $C(s,a)$, the effective reward is $\widetilde{R}_c = (1-\epsilon)R + \epsilon C$. With standard Robbins–Monro stepsizes, vanilla Q-learning converges to the fixed point $\widetilde{Q}_c^\star$ of the resulting perturbed Bellman operator. The five-state example gives, for $0 < d \le \bar{r}$ and any $\kappa > 0$,

$$C = \frac{(2-\epsilon)d + \kappa}{\epsilon} \quad \Longrightarrow \quad \|\widetilde{Q}_c^\star - Q^\star\|_\infty = 2d + \kappa, \tag{1}$$

with the greedy action at the initial state also reversed.

2. Finite-time guarantee for the robust algorithm. For $\delta \in (0,1)$ and $\alpha = \log T/[(1-\gamma)T]$, the robust output satisfies, with probability at least $1-\delta$,

$$e_T \le \frac{e_0}{T} + \mathcal{O}\left(\frac{\bar{r}\log T}{(1-\gamma)^{5/2}}\sqrt{\frac{\log(SAT/\delta)}{T}} + \frac{\bar{r}\sqrt{\epsilon}}{1-\gamma}\right). \tag{2}$$

The clean-data term matches the usual $\widetilde{\mathcal{O}}(T^{-1/2})$ rate; corruption contributes an additive term independent of attack magnitude.

3. Dynamic thresholding controls rare estimation failures. Let $T_{\mathrm{lim}} := \lceil 2\log(4/\delta_1)\rceil$. The clipping radius is

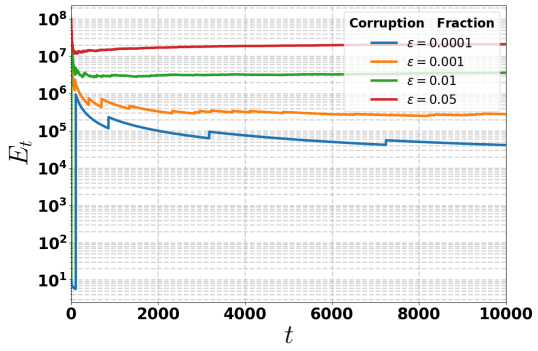
$$G_t = \begin{cases} 2\bar{r}, & 0 \le t \le T_{\mathrm{lim}}, \\ C\bar{r}\big(\sqrt{\log(4/\delta_1)/t} + \sqrt{\epsilon}\big) + \bar{r}, & t \ge T_{\mathrm{lim}} + 1. \end{cases} \tag{3}$$

Early clipping supplies deterministic control before the robust estimator is accurate; later, the threshold tracks its statistical confidence radius.

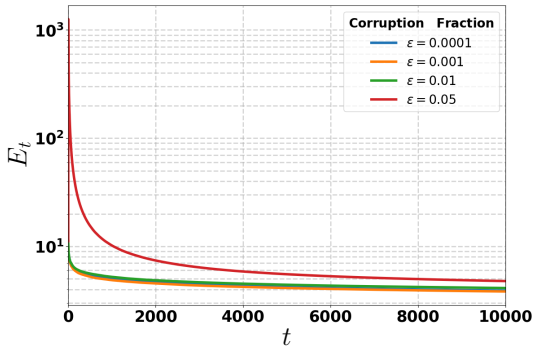
Theoretical comparison.

	Vanilla synchronous Q-learning	ϵ-Robust Q-Learning
Attack sensitivity	Error can be arbitrarily large	Bound independent of attack magnitude
Clean statistical rate	$\widetilde{\mathcal{O}}(T^{-1/2})$	$\widetilde{\mathcal{O}}(T^{-1/2})$
Robustness mechanism	Uses each observed reward directly	Historical filtering and adaptive clipping

Numerical results. Plots show $E_t = \|Q_t - Q^\star\|_\infty$ for $\epsilon \in \{0.0001, 0.001, 0.01, 0.05\}$, reward variance $\sigma^2 = 10$, and a 10^5 reward-biasing attack; all curves average 100 independent runs.



(a) Vanilla-Q under reward corruption.



(b) Robust Sync-Q under the same attack.